MARINE CORPS SYSTEMS COMMAND

Condition Based Maintenance (CBM+)

Virtual Industry Day

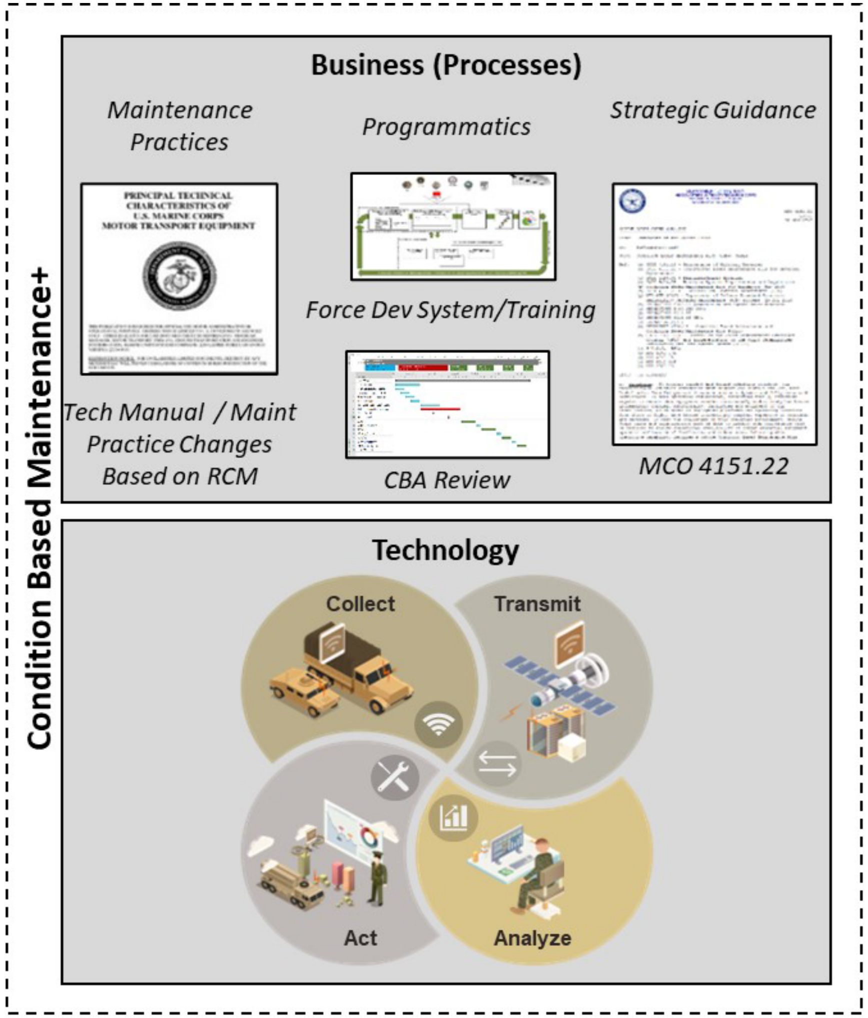

OUR MISSION
Equip our Marines with the most capable ground warfare and information technology systems to maximize their expeditionary readiness and combat effectiveness in all domains.
OUR VISION
Equipping an unrivaled, future-focused Fleet Marine Force, powered by a dedicated acquisition workforce.

USMC CBM+ VISION

*CBM+ is a **shift in maintenance behavior** and practices enabled by technology*

Current Maintenance Posture

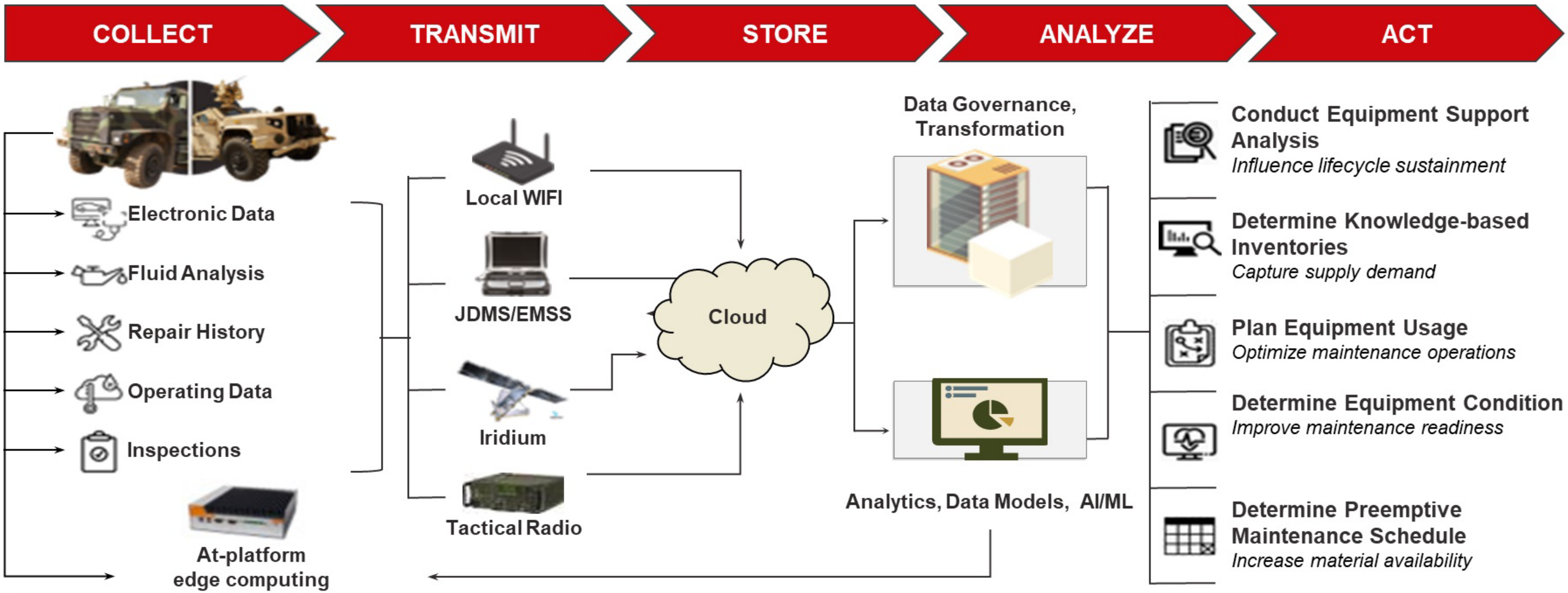
- **Limited ability to conduct just 2 condition monitoring elements** (inspections and repair history data)
- Performed **reactively** when there are **failures**
- Maintenance based on **time**
- Diagnostics data stuck at weapon system, **cannot be offboarded**
- **Limited, manual** failure analysis capabilities
- **Unpredictable** future operational availability



Future Maintenance Posture

- **Full use of all 5 elements of condition monitoring** (fluid analysis, repair history, inspections, electronic data, and site conditions)
- **Reliability Centered Maintenance** analysis feeds weapon systems' CBM+ Strategies
- Performed **proactively,** based on **predicting remaining useful life**
- Maintenance based on **anticipated events**, performed only as needed
- Diagnostics successfully **offboarded, centralized, and analyzed**
- Failure analysis performed in an **automated manner**
- **Data-driven insight** on future operational availability

CBM+ CONCEPTUAL FRAMEWORK



CBM+ is the application technologies that can be implemented now as well as yet-to-be developed capabilities and it is a system that changes how Marines **act** in response to equipment data that is **collected**, **transmitted**, **stored**, and **analyzed**.

USMC CBM+ IMPLEMENTATION

"We will make strategic investment in **Data Science**, **Machine Learning** and **Artificial Intelligence**. Initial investments will be focused on challenges we are confronting in talent management, **Predictive Maintenance**, Logistics, intelligence, and training." – 38TH Commandant's Planning Guidance

Up to FY19

Individual Pilots

Conduct pilots on individual weapon systems and gather initial learnings

Individualized Pilots

LAV

M777

M88

- Gauged impact from use of sensors and changes to scheduled maintenance
- Conducted 7 total CBM+ related pilots
- Focused on individual functional capabilities of CBM+ (collect, transmit, store, analyze, act)

FY20

CBM+ MVP Started

Build a full-spectrum, tightly scoped CBM+ Minimum Viable Product (MVP)

Minimum Viable Product

JLTV

MTVR

- Enabled a total of 10 JLTVs and 10 MTVRs
- Partnered with 1 CONUS unit
- Leveraged initial learnings from existing pilot efforts
- Synchronized efforts across spectrum of stakeholders

FY21-24

CBM+ Expansion

Evaluate and expand MVP to additional units

MVP Expansion

JLTV

MTVR

- Expanded JLTV/MTVR MVP to multiple units
- Established a CBM+ cloud solution within Jupiter/Advana
- Measured impacts on *material readiness,* and decision support
- Expanded to include LVSR, TRAM, MCT, HYEX

FY25-29

CBM+ Scaling

Scale aggressively to Marine Logistics Groups, additional weapon systems, and new acquisitions with POM-26 FYDP

Scaling CBM+ Fleetwide

MHE

MADIS/LMADIS

ACV

- Installed on LMADIS and MADIS platforms
- Building an interface to GCSS-MC
- Planned installs at AVTB summer of 2026

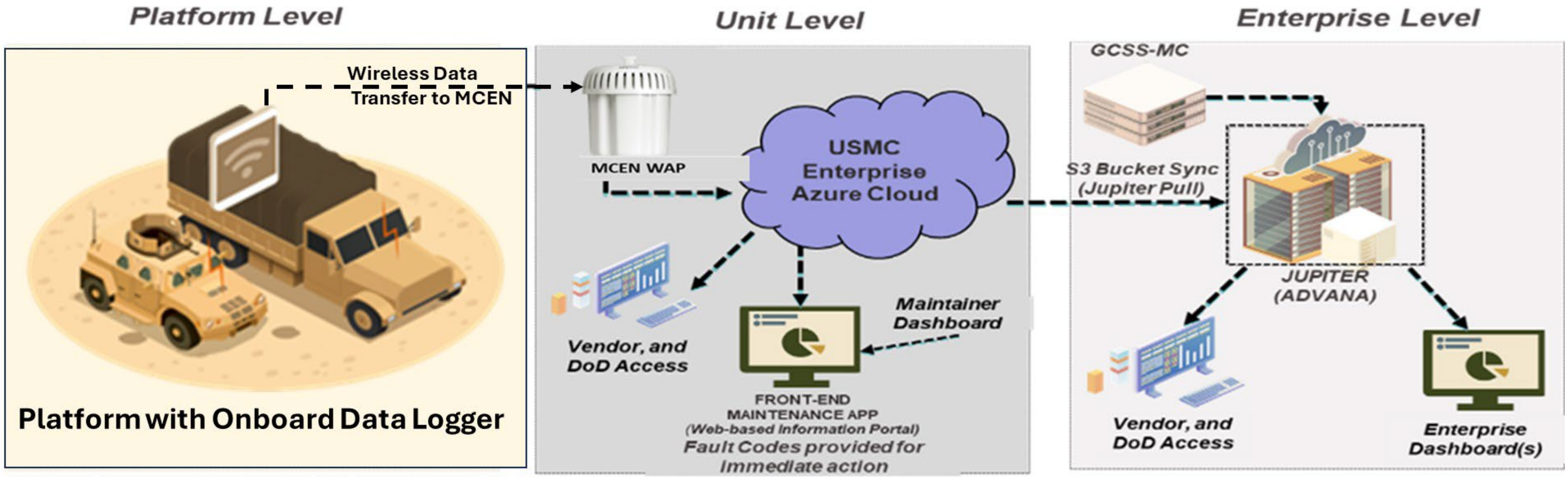
FY30

Realized CBM+ Value Force Design 2030

Global Logistics Awareness will support the **lethality** of Marine Corps forces by **enabling decision** and execution **superiority**, allowing commanders to **outpace** the enemies' **decision** cycle. It will do so by allowing the **logistics enterprise** to deliver the *right* resources, to the *right* place, at the *right* time, for the *right* reasons.

> *Sustaining the Force for the 21st Century*
> *- Commandant of Marine Corps*

USMC CBM+ IMPLEMENTATION



Initial Target Platforms

MTVR
JLTV
LVSR
MCT
TRAM
Excavator

- *Aggregated data is processed and decoded in* ***Azure on the MCEN*** *providing maintainer dashboard for immediate feedback*
- *Use existing MCEN Wireless Access Points (WAPS)*

- *DoD and Vendor users have access to data files and GCSS-MC data in Jupiter.*
- *Data Analytics, Artificial Intelligence, and Machine Learning applied to the data sets on the Jupiter Platform*
- *Web-Based Enterprise Health Dashboard(s) available to the FMF*

WHAT WE ARE SEEKING FROM INDUSTRY

- A single contract solution for hardware, data analytics, and cyber aspects of the program. Can be achieved through subcontracting/partnerships

- Software/Data analytic solutions that are not proprietary. The USMC intends to own not only the data, but the models and visualizations created by this contract

- Data logger hardware solution(s) that are TRL-9 and capable of WI-FI at a minimum. Multiple data transmission methods (Wi-Fi, Iridium, Cellular, LI-FI, etc.) is desirable

- Demonstrated experience building models for ground vehicles

- Ability and willingness to work in Government provided cloud solutions (MS Azure and Advana)

- Data visualizations that provide Marine Commanders, Staff Officers, and Maintainers a decision support tool that is easy to use and understand


QUESTIONS?


EQUIPPING OUR
MARINE CORPS SYSTEMS COMMAND
SEMPER FIDELIS
EQUIPPING OUR MARINES
MARINES